

Slingshot Space Launch Graphics Concepts V2.0

Engineering Visualization Brief: 15 High-Fidelity Graphic Design Prompts

Version 2.0 — Revised & Expanded | June 2026

Δ Change Log

- Updated all graphic prompts to 2025 ANSYS Fluent / SolidWorks 2025 rendering aesthetic
- Added V2.0 prompts (16–18) for CNT actuator concepts and orbital trajectory analysis
- Color palette standardized: dark navy (#1B2631) background, white labels, cyan (#00FFFF) accent
- Cross-reference to V2.0 updated performance projections (273 m/s vs previous estimates)

Style Guide (All 18 Graphics)

Aesthetic: Clean engineering schematic. Dark navy background (#1B2631). White line drawings. Cyan (#00FFFF) and YInMn Blue (#1B4F72) accent colors. Labels in Helvetica Neue or Noto Sans. Dimensions in metric SI units. Grid at 1/10 of main dimension scale.

Part I — Comparative Actuator Schematics

Graphic 1 — Dielectric Elastomer Operating Principle Two-panel “before/after” cross-section of a DE actuator. Left: 0V state — thick elastomer layer (yellow) between two compliant electrode plates (dark grey). Right: High-V state — elastomer compressed thin (Maxwell stress), length expanded 40%. Arrow showing payload direction upward. Labels: “Maxwell stress”, “Electrode displacement”, “Strain = 40%”. Caption: “DE Actuator: Voltage compresses □ lateral expansion □ linear contraction of rope.”

Graphic 2 — SMA vs. DE Energy Budget Bar Chart Horizontal bar chart showing: DE energy output (3.4 kJ/kg, solid blue bar), SMA required heating energy (75 kJ/kg, red bar extending far right), SMA mechanical output (6 kJ/kg, small orange bar). Title: “Why SMA Fails at Altitude: Thermal Overhead.” Grid in kJ/kg. Legend box upper right.

Graphic 3 — CNT Yarn Twist-Actuated Rope Isometric view of braided multi-wall CNT yarn. Twisted coil structure with arrows showing electrical current direction inducing torsional rotation. Inset: molecular diagram of MWCNT with π -electron cloud highlighted. Energy density label: “31 kJ/kg demonstrated.”

Part II — Environmental Failure Diagnostics

Graphic 4 — Altitude Environment Gauntlet Vertical cross-section of atmosphere from ground to 35 km. Temperature profile (red line, $\approx 60^\circ\text{C}$ at 30 km). Pressure profile (blue

line, 0.001 atm at 30 km). UV flux intensity bar (increasing with altitude). Zones labeled: “Troposphere”, “Tropopause”, “Stratosphere”. Balloon position marked with icon at 30 km.

Graphic 5 — DE Vacuum Collapse Failure Mode Two-panel comparison. Left: Air-filled DE at sea level — round uniform cells. Right: Same DE at 0.001 atm — cells collapsed inward, void spaces, elongated stress concentrations marked in red. Caption: “Solution: N₂ pre-fill prevents collapse.”

Part III — Final System Architecture

Graphic 6 — Complete System Assembly Blueprint Exploded view isometric: High-altitude balloon (top), 50 m suspension tether, capacitor discharge module, 400 m DE rope coil (in N₂-purged housing), payload capsule (2 kg sphere). All components labeled with mass and material. Total system mass summary box.

Graphic 7 — Rope Deployment Sequence Storyboard 6-frame horizontal sequence: (1) balloon at altitude, rope coiled; (2) capacitor charge confirmation; (3) trigger discharge — rope begins contracting; (4) peak contraction — payload accelerating; (5) rope fully contracted — payload at 273 m/s; (6) payload coasting on ballistic arc to apogee. Time stamps under each frame.

Graphic 8 — Velocity vs. Launch Altitude Comparison Line chart. X-axis: altitude (0–35 km). Y-axis: payload velocity (0–400 m/s). Three curves: sea level (lowest, ~80 m/s due to drag), 20 km (~180 m/s), 30 km (~273 m/s). Kármán line velocity requirement (7,800 m/s) shown as red dashed horizontal — indicating significant gap.

Graphic 9 — Power Distribution Sankey Diagram Sankey flow diagram: Input energy 183,600 J (blue, left). Splits: rope kinetic 42% □ 77,112 J; rope losses 28% □ 51,408 J; thermal dissipation 18% □ 33,048 J; payload 12% □ 22,032 J. Payload kinetic = $\frac{1}{2} \times 2 \times 273^2 = 74,529$ J at 40.6% efficiency. Updated with V2.0 numbers.

Part IV — V2.0 New Graphics

Graphic 16 — CNT Yarn Scale-Up Roadmap Timeline chart (2024–2032). Current state: 1 m CNT yarn demonstrated. Milestones: 10 m (2026 estimated), 100 m (2028), 400 m production rope (2031). Energy density maintained projection (31 kJ/kg). If achieved: payload velocity jumps to 1,200 m/s — reaching lower sub-orbital regime.

Graphic 17 — Two-Stage Launch Architecture Trajectory diagram showing: (1) DE rope impulse to 273 m/s at 30 km, (2) solid rocket second stage fires at apogee (80 km), (3) final orbital velocity 7,800 m/s. Delta-V budget bar: DE stage 273 m/s + rocket stage 7,527 m/s. Underscores the rocket complement requirement.

Graphic 18 — Safety Abort Sequence Flow diagram: nominal launch (green path) vs. abort scenarios. Abort triggers: voltage anomaly, weather event, balloon drift. Abort actions: N₂ vent (disarms rope), pyrotechnic separation, balloon climb to safe altitude. All abort paths end at “Balloon retrieval” green box.

